

Price The Option - 1

To accurately determine the price for the down-and-out barrier call option, the Milstein scheme was implemented to solve the SDE:

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (1)$$

The Milstein scheme is strong and weakly convergent; therefore it is a suitable method to use when solving for the option price of a path-dependent option like a down-and-out barrier call option:

$$\tilde{X}_{n+1} = \tilde{X}_n + a(\tilde{X}_n) \delta t + b(\tilde{X}_n) \Delta W_n + \frac{1}{2} b'(\tilde{X}_n) b(\tilde{X}_n) ((\Delta W_n)^2 - \delta t) \quad (2)$$

$$\text{Where: } \tilde{X}_t = S_t \quad a(\tilde{X}_n) = rS_t \quad b(\tilde{X}_n) = \sigma S_t$$

We evaluate ΔW_n through the properties of a white noise process. The increment $\Delta W_n = W_{t_{n+1}} - W_{t_n}$ is distributed as $\Delta W_n \sim N(0, \delta t)$, which can be approximated as $\Delta W_n = \sqrt{\delta t} Z$, where $Z \sim N(0, 1)$. The standard normal part, Z is generated in Python using `np.random.normal()`. Similarly, we evaluate the term $((\Delta W_n)^2 - \delta t)$ by using $(Z^2 - 1) \delta t$, where Z is again generated using `np.random.normal()`.

The Milstein Scheme option price, a_M and the standard deviation b_M was evaluated first using 10000 monte carlo simulations, from this we found the 95% confidence interval where 1.96, from equation (3), corresponds to the standard normal value of the 5% tail of the standard normal distribution.

$$\left[a_M - 1.96 \frac{b_M}{\sqrt{M}}, a_M + 1.96 \frac{b_M}{\sqrt{M}} \right] \quad (3)$$

Equation (3) shows that increasing the number of monte carlo simulations M causes the interval width to decrease by a factor of $\frac{1}{\sqrt{M}}$. To determine the number of Monte-Carlo simulations that give a CI width of \$0.01 we can re-arrange for M setting 0.01 as this width to get:

$$M = (392b_m)^2 \quad (4)$$

The number of samples required to make the confidence interval equal to \$0.01 first required finding the computational time at $M = 10000$ Monte-Carlo simulations. Then scaling the computational time at this value according to equation (4), gives the required amount of Monte-Carlo simulations to reduce the confidence interval to \$0.01. We will investigate later how this method of reducing the confidence interval, by increasing M is inefficient, and the confidence interval can be reduced by using other techniques.

The computational time for $M = 10000$ was found by running the Milstein-Scheme function over a for loop that calculates the time taken for the function to run by using `time.perf_counter()`. Because of the single vectorised for loop we can extrapolate that the computational time increases linearly with $O(M)$, meaning that the base computation time (the computational time required to make the CI a width of approximately \$0.01) scales with M according to:

$$\left(\frac{M \text{ required for CI} = 0.01}{10000} \right) \times \text{Computational time for } M = 10000$$

Using the theory described in this part the various metrics calculated for the down and out call option price are shown in the table below:

Metric	Value
Down and Out Option Price	12.283
Confidence Interval (95 %)	[12.071, 12.496]
Width of Confidence Interval	0.426
Standard Deviation	10.858
Mean Computational Time at M = 10000	0.191 s
Number of Simulations for CI Width = \$0.01	18 116 000
Computational Time for CI Width = \$0.01	345.607 s

Table 1: Down and Out Option value parameters calculated using the Milstein-scheme.

Variance Reduction: Antithetic Variates - 2

Using antithetic variates we can generate two samples of standard normal variates:

$$\gamma = \{Z_0, Z_1, \dots, Z_{N-1}\} \quad \text{and} \quad -\gamma = \{-Z_0, -Z_1, \dots, -Z_{N-1}\}$$

These sequences were used to compute the option price by averaging the two M number of terminal option prices generated by the two Monte-Carlo processes.

$$Y = \frac{1}{2}(f(\gamma) + f(-\gamma))$$

The benefit of using antithetic variates comes from the two sequences, $f(\gamma)$ and $f(-\gamma)$ (being stock and option prices) that are negatively correlated. Because of the dependence between the two sequences, the overall variance of the predicted prices should be lower, because the covariance is negative, this is seen with the variance of dependent variables:

$$\text{Var}(Y) = \frac{1}{4}(\text{Var}(f(\gamma)) + 2 \text{Cov}(f(\gamma), f(-\gamma)) + \text{Var}(f(-\gamma))) \leq \frac{1}{2}\text{Var}(f(\gamma)).$$

$\text{Cov}(f(\gamma), f(-\gamma))$ is negative therefore we therefore expect a lower confidence interval, width and standard deviation of the predictions for the stock and option prices.

The correlation between the terminal stock prices generated by the two sequences was -0.98546, and the correlation between the terminal down and out option prices generated by the two sequences was -0.85628. This is what we should expect for the antithetic variate samples, because they are generated from the same standard normal sample. The correlation between the antithetic and regular stock price is lower than that calculated from the option prices, this is because calculating the down and out option price introduced the boundry condition that gives the option price a value of zero if the stock drops below the specified boundry at any time. This likely made the option prices from the two samples slightly more correlated, which caused the antithetic option price to be slightly more correlated than the antithetic stock price estimate. The metrics shown in the previous two parts were then calculated and are illustrated in the table below:

Metric	Value
Down and Out Option Price	12.352
Confidence Interval (95 %)	[12.295, 12.410]
Width of Confidence Interval	0.115
Standard Deviation	2.931
Mean Computational Time at M = 10000	0.221 s
Number of Simulations for CI Width = \$0.01	1 320 100
Computational Time for CI Width = \$0.01	29.192 s

Table 2: Down and Out Option value parameters calculated using antithetic variates.

The confidence interval width for the option price was much smaller due to the reduction in the standard deviation. The computational time for the function to run was larger because the antithetic estimate method required running the Monte-Carlo simulation for two sequences: $f(\gamma)$ and $f(-\gamma)$ rather than just one. The overall effect of the antithetic variate's reduction in the variance at $M = 10000$, meant that the base computational time was over 10 times smaller than the base computation time calculated in part 1.

Variance Reduction: Fine-Tuned Control Variates - 3

In this part we will use a control variate, S to get an adjusted estimate for V_θ so that the variance of our estimated option and stock price is small. This control variate estimate is given by Z_θ :

$$Z_\theta = V + \theta(\mathbb{E}[S] - S) \quad (5)$$

θ was chosen to minimise the variance of Z_θ given by the quadratic equation:

$$\text{Var}(Z_\theta) = \text{Var}(V) - 2\theta\text{Cov}(V, S) + \theta^2\text{Var}(S)$$

To find an optimal value for θ , the variance of Z_θ must be minimised, by differentiating $\text{Var}(Z_\theta)$ with respect to θ , and setting $\frac{d\text{Var}(Z_\theta)}{d\theta}$ to zero solves for the value of θ that gives minimum variance:

$$\theta_{\min} = \frac{\text{Cov}(V, S)}{\text{Var}(S)} \quad (6)$$

The value of theta that minimises the variance of the option price is illustrated below:

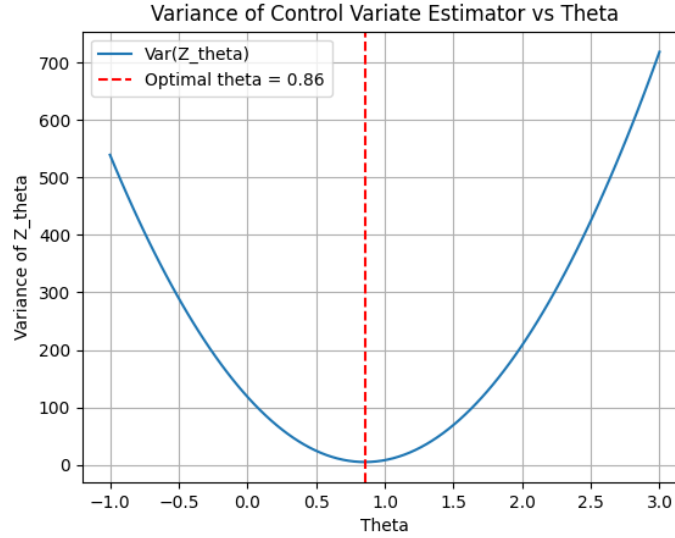


Figure 1: Variance of Z_θ plotted against a range of values for θ , with the optimised value shown in red.

The plot shows the value of θ that minimises the variance by the dotted red line around 0.86. To find the base computation time we can implement a similar approach used to find the computational time in the previous two parts. Firstly by calculating the computational time taken to find the optimised θ using equation (6); then by calculating the computational time for the actual control variate estimate Z_θ , which was implemented in a separate function to the optimised θ (so that we could investigate how different values of θ changed the variance of Z_θ).

Calculating the covariance, from equation (6), between the option value and stock price generates the majority of the computational complexity when finding the optimised value for θ , this complexity scales as $O(M)$ where M is the number of Monte-Carlo simulations. Similarly the computational time for the control variate estimate scaled as $O(M)$, and likewise for the original Milstein-Scheme function used in part 1. A for loop was used again to generate the mean time from a list of times taken to find the optimised θ and the the control variate estimate, both of which were added onto the mean time taken for the Milstein-Scheme function to run in part 1. Since all three components scale as $O(M)$ the base computation time was calculated in the same way as the previous two parts. The metrics calculated by the control variate estimate are tabulated below:

Metric	Value
Down and Out Option Price	12.376
Confidence Interval (95 %)	[12.330, 12.423]
Width of Confidence Interval	0.094
Standard Deviation	2.388
Mean Computational Time at $M = 10000$	0.191 s
Number of Simulations for CI Width = \$0.01	876 260
Computational Time for CI Width = \$0.01	16.7310 s

Table 3: Down and Out Call Option terminal time value parameters calculated using the control variate estimate.

Because the control variate estimate reduces the variance at a given level of M , the required Monte-Carlo simulations is much smaller than the estimates from the previous two parts. Ad-

ditionally the base computation time calculated using the control variate estimate is smaller than the computational time required in part 2 because the function uses the original Milstein-Scheme function, that utilises only 1 set of Monte-Carlo calculations, rather than averaging over two individual paths. Since the computational time and the confidence interval is smaller at $M = 10000$, the base computation time is much smaller.

Variance reduction: antithetic and fine-tuned control variates - 4

To implement the control variate first and the antithetic variate after we need to calculate theta using `V_plus`, `S_plus` and `V_minus`, `S_plus` from equation (6). Using these values of theta we estimate the control variate estimate, Z_θ in the same way as part 3. Using the two sets of paths Z_θ^+ and Z_θ^- we calculate the antithetic variate estimate by averaging the paths and finding the overall mean in same way as done in part 2. The other metrics were calculated in the same way as the previous parts to give the results:

Metric	Value
Down and Out Option Price	12.346
Confidence Interval (95 %)	[12.306, 12.386]
Width of Confidence Interval	0.080
Standard Deviation	2.031
Mean Computational Time at $M = 10000$	0.221 s
Number of Simulations for CI Width = \$0.01	634 130
Computational Time for CI Width = \$0.01	14.033 s

Table 4: Down and Out Call Option terminal time value parameters calculated using the control variate then antithetic variate estimate.

To implement the antithetic variate first and the control variate after we calculate theta using equation (6) by averaging the paths `V_plus` and `V_minus`; likewise for `S_plus` and `S_plus`. Using this averaged path we find the value of theta and estimate the control variate estimate, Z_θ in the same way as part 3 which give the results shown below:

Metric	Value
Down and Out Option Price	12.332
Confidence Interval (95 %)	[12.325, 12.340]
Width of Confidence Interval	0.015
Standard Deviation	0.389
Number of Simulations for CI Width = \$0.01	23 273
Computational Time for CI Width = \$0.01	0.515 s

Table 5: Down and Out Call Option terminal time value parameters calculated using the antithetic variate then control variate estimate.

Using the antithetic variate first and the control variate after gave a much smaller confidence interval than doing the control variate first and the antithetic variate after. This is because the values of theta generated by `V_plus`, `S_plus` and `V_minus`, `S_plus` is optimised to find Z_θ^+ and Z_θ^- individually, and combining the two estimates after don't actually represent an optimal choice for the control variate estimate. Conversely using the antithetic variates first, an accurate set of option prices is transformed under the control variate estimate, to give a value of theta that minimises the overall variance.

The computational time required to calculate the option price using antithetic variates and fine-tuned control variates was only about a factor of 0.0001 seconds slower than the time taken to compute for antithetic variates, which is too small to show on the table. Therefore recalling the individual functions again to test computational time showed that it sometimes wasn't slower, due to the randomness in the `time.perf_counter()` function. Because there is added computational complexity by calculating the control variate estimate, this method is marginally slower than the antithetic variate estimate. Therefore we used the computational times from the individual control variate, antithetic and covariance calculations to remove randomness and find the antithetic and fine-tuned control variate computational time.

Results and Discussion - 5

The model which produced the lowest base computation time for achieving a confidence interval equal to \$0.01 was the antithetic then control variate estimate. Because the base computation time for the antithetic then control variate estimate was approximately 36 000 000 times smaller than the original Milstein-Scheme function it was easily found, giving a value of: \$12.335. All of the implementations of the Milstein-Scheme and methods for reduction in variance have this accurate estimate within their confidence interval, but using the antithetic then the control variate yielded the most accurate results. Using the control variate first and then the antithetic variate to estimate the option price was slightly more efficient than just using the control variate. Finally whilst the antithetic variate estimate was over 10 times more efficient in obtaining a confidence interval of 0.01 than the original Milstein-Scheme, it took almost twice the base computation time compared to the fine-tuned control variate.



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